

Exploring the Path Loss of a Hacking Tool for Security Matters in the Internet of Things

Roxana Mata-Hernandez
Faculty of Science
Autonomous University of San Luis Potosi (UASLP)
San Luis Potosi, Mexico
a264833@alumnos.uaslp.mx

Marco Cardenas-Juarez
Faculty of Science
UASLP
San Luis Potosi, Mexico
marco.cardenas@uaslp.mx

Jorge Simón
Cátedras CONAHCYT
CONAHCYT
Zacatecas, Mexico
jsimonro@conahcyt.mx

Enrique Stevens-Navarro
Faculty of Science
UASLP
San Luis Potosi, Mexico
enrique.stevens@uaslp.mx

Alessandra Rizzardi
Department of Theoretical and Applied Science (DiSTA)
University of Insubria
Varese (VA), Italy
alessandra.rizzardi@uninsubria.it

Abstract—The rapid expansion of the Internet of Things (IoT), now with billions of mainly wireless interconnected devices, has brought concerns and challenges regarding the security of IoT networks and devices, as they are often vulnerable to attacks. In this article, a measurement campaign is carried out to investigate the path loss experienced by a portable IoT hacking tool, known commercially as *Flipper Zero*TM, when used in transmitter mode, to shed light on the maximum achievable distance at which an IoT device can still receive an eavesdropper's signal above a minimum power level. The path loss measurements are performed in three different outdoor environments. The results show that the hacking tool transmitted signal can reach up to 15 meters with power above -90 dBm, which is still in the sensitivity range of many IoT devices, thus revealing potential vulnerabilities and security risks in real-world scenarios.

I. INTRODUCTION

The Internet of Things (IoT) has the potential to improve diverse aspects of our lives; hence, we have witnessed its exponential growth in recent years, with billions of interconnected devices, mainly in a wireless manner. Indeed, the IoT has transformed vertical markets such as cities, agriculture, water, security & emergencies, retail, industrial control, home automation, and e-health, amongst others, enabling the deployment of smart applications, thus, the economic growth of IoT-based services is considerable for such businesses [1]. However, this rapid expansion has also introduced several challenges in terms of security and privacy [2], mainly due to the wireless nature of communications. To cope with such issues, a large number of methods and solutions have been recently proposed, for example, adopting machine learning and deep learning techniques [3], [4]. Moreover, to ensure the integration and interoperability of IoT networks and devices, it is crucial to address vulnerabilities at every layer of the IoT architecture, as presented in [5].

The authors would like to express their gratitude to the Mexican National Council for Humanities, Science, and Technology (CONAHCYT) for supporting this project through the master's thesis funding of Roxana Mata Hernandez, CVU number: 1240167

Particularly, the physical (PHY) layer, which encompasses the physical transmission of data, including radio frequency (RF) signals, modulation techniques, and the measurement of signal strength, serves as the foundation for communication in IoT networks. Since the PHY layer is the first point of contact between IoT devices, it is particularly vulnerable to potential security threats, making it a critical area of research. Furthermore, the appearance and commercial availability of different gadgets and hacking tools, either in software or hardware, increases the risks, making security more vulnerable in IoT networks. For example, in [6] the signals from a wireless electric switch are analyzed using a software-defined radio operating in the 433 MHz license-free industrial, scientific, and medical (ISM) frequency band. In this experiment, the author obtained the ON and OFF symbol bursts of the wireless electric switch, which can be replicated and transmitted to hack the wireless device, thus exhibiting a vulnerability of the system.

In this article, we focus on analyzing the large-scale path loss experienced by the signals transmitted from a portable hacking tool, commercially known as *Flipper Zero*TM¹. On the one hand, *Flipper Zero*TM is a versatile and widely used hacking device known for its capabilities to explore the security of diverse wireless technologies, such as near-field communication (NFC), radio frequency identification (RFID), Bluetooth, Infrared, and others below 1 GHz. Particularly, the frequencies below 1 GHz are the operating range of access control systems, remote keyless systems, and IoT sensors. This hacking tool can be used to capture and analyze radio frequency signals, allowing the execution of an eavesdropping attack. Such attacks exploit the vulnerabilities of poorly secured wireless transmissions and steal information from IoT devices [2]. Then, the captured information can be transmitted by the hacking tool to access or take control of a wireless

¹The official website of Flipper Zero is <https://flipperzero.one/>

system in the IoT context. On the other hand, the path loss is a crucial factor that affects the reliable transmission of signals. By performing a series of path loss measurements, we aim to shed light on the potential vulnerabilities and security concerns on IoT devices, that might arise when employing Flipper Zero™ in transmitter mode, since path loss directly impacts the distance range from which the hacking tool can affect the reliability and security of IoT devices. The present study assesses three different scenarios in which IoT users and devices might operate: (i) line-of-sight, (ii) crowded area (e.g. student area of a University), and (iii) very busy street. This evaluation helps in assessing whether an attacker could compromise the security of a device by being in close proximity and intercepting the signals.

The rest of the article is organized as follows: Section II presents security issues in the physical layer of the IoT context. The path loss measurement methodology is detailed in Section III. The obtained results are shown in Section IV. Finally, Section V presents the conclusions of the study and draws some hints for future research directions.

II. IOT SECURITY ISSUES IN THE PHYSICAL LAYER

The rapid growth of IoT networks has also introduced significant security challenges, particularly, in the physical layer [7]. Some of the main security issues that can be found in IoT wireless networks are the following:

- *Eavesdropping*. This is the most common type of attack in the PHY layer. An attacker can eavesdrop on the wireless communication between two IoT devices by simply listening to the radio waves, which can be accomplished using either a basic radio transmitter or more advanced equipment.
- *Jamming*. This is another common type of attack in the PHY layer. An attacker can jam the wireless communication between two IoT devices by transmitting a signal that interferes with the legitimate signal. This can be done with a simple radio transmitter, or with more sophisticated equipment.
- *Spoofing*. This is a more sophisticated attack in the PHY layer. An attacker can spoof the identity of an IoT device by transmitting a signal that appears to come from a legitimate device. This can be used to gain access to the network or to disrupt the network.
- *Contamination*. This is a type of attack in which the attacker corrupts the wireless communication between two IoT devices. This can be done by transmitting a signal that contains noise or other interference. This can be used to disrupt the network or to corrupt the data that is being transmitted.
- *Physical tampering*: This is an attack in which the attacker physically tampers with an IoT device to gain access to it. This can be done by opening the device, replacing components, or adding malicious code.

In recent years several hacking gadgets and tools have been made commercially available. Among them, Flipper Zero™ is a portable hacking tool that can be used to scan, analyze,

and interact with a variety of wireless devices. Furthermore, the tool is handheld and easy to carry around, open-source, and new features can be added by developers. The reach of hacking tools into the common market has allowed the rise of IoT attacks where even people with bare technical skills can become a serious threat if preventive measures are not taken. The focus of this study is on the sub-1 GHz transceiver of the device to explore the vulnerabilities in IoT sensors by considering the actual achievable distance of the device's transmitter under different fading conditions.

In the IoT context, there are different devices with different receivers' sensitivity below which the device will not operate. Although, the sensitivity will vary according to the standard and technology [8]. In Table I we summarize some reference sensitivity values for common IoT devices. It is worth it to remark that the actual minimum received signal power may vary depending on the specific device and the environment in which it is operating as well as the distance between the device and the base station or gateway, the noise power in the environment, the modulation scheme used by the IoT device, and the signal bandwidth. It is important to consider the minimum received signal power above which an IoT device will operate correctly. If the signal power is too low, the device may not be able to communicate and process the signal's information. Therefore, it is interesting to know what is the maximum distance from which a transmitted signal from a hacking device will arrive at an IoT device with power above this threshold (i.e. receiver's sensitivity).

TABLE I
COMMON IOT DEVICES AND THE MINIMUM RECEIVED SIGNAL POWER
THEY NEED TO OPERATE

Device	Technology	Minimum received signal power (dBm)
Smart thermostat	Wi-Fi	-85
Smart light bulb	Zigbee	-90
Air quality sensor	LoRa	-120
Wearable fitness tracker	Bluetooth	-80
Smart meter	NB-IoT	-130
Connected car	LTE-M	-100
Medical implant	Wi-Fi	-95

III. PATH LOSS MEASUREMENT METHODOLOGY

The Flipper Zero's™ sub-GHz transceiver operates in the 300-348 MHz, 387-464 MHz, and 779-928 MHz frequency bands. The availability of each range is also restrained within each country's legislation. This module contains a C1101 chip and an antenna with a maximum working range of 50 meters when operating at the highest transmission power. Moreover, the output power is programmable, and the full range is available in all frequency bands. Further experimental measurements showed that the default output power of the transmitter was the lowest setting of -30 dBm, which is the selected output power in this study. The sub-GHz module supports functions such as spectrum analyzer, reading and decoding common RF signals, reading and saving raw RF signals, and transmitting saved RF signals. IoT devices like smart

lights, wireless doorbells, garage doors, and many others work within previously mentioned frequency ranges. Therefore, these IoT devices are potentially vulnerable to eavesdropping attacks using a device like Flipper Zero™, since signals can be captured, analyzed, and transmitted by an eavesdropper to reach a targeted IoT device to hack it. To quantify the maximum distance from which an eavesdropper can reach an IoT device using Flipper Zero™, the path loss experienced by a Flipper Zero™ transmitted signal is studied in three different scenarios with the default transmission power, as described next.

A. Measurement scenarios

To determine the distance, from which Flipper Zero™ sub-GHz transmitter can interact with IoT receivers, three outdoor scenarios are taken into consideration: (i) a line-of-sight, (ii) a crowded student area inside the Autonomous University of San Luis Potosi campus (Mexico), and (iii) a very busy street with highly dynamic behavior. For the first scenario, where a static environment with a direct line of sight is needed, a building roof was chosen, as shown in Fig. 1. This scenario is used to set a baseline on how the transmitted signal fades in the air in the absence of other significant components such as walls, trees, and others. The second scenario takes place in a student area with crowded traffic of people walking by inside a campus, as shown in Fig. 2. The third scenario is a sloping street with heavy vehicular traffic, as shown in Fig. 3.

B. Experimental setup and data collection

We consider that an attack can be performed in any of the investigated scenarios. In the experiments, it is assumed that the eavesdropper captures and analyzes a signal from an IoT device and it is ready to transmit it to reach this target. To emulate this signal, we use a pure tone at 925 MHz from an RF signal generator. Then the signal is recorded into the Flipper Zero™ with the *Save Raw* option [9]. Note that the emulated eavesdropper signal is going to be transmitted by the Flipper Zero™ and the received power at different distances is going to be measured to study the path loss. Therefore, the duration of this signal is recommended to last enough to capture different power samples at the receiver (i.e. the IoT device location). For the sake of taking the measurements with more simplicity, in this experiment, we recorded a signal with a duration of 6 minutes, approximately, so we could move away from the Flipper Zero™ transmitter and record the power measurements at different locations. Then, for each of the studied scenarios, Flipper Zero™ transmits a pure tone signal (i.e. emulated eavesdropper signal) with the lowest transmission power (i.e. -30 dBm) of the CC1101 transceiver [10]. Although the theoretical output power from Flipper Zero™ sub-GHz transmitter should be of -30 dBm, experimental measurements found that, in practice, it is about -34 dBm. On the receiver side, a measuring station using a spectrum analyzer and a log-periodic antenna is placed to measure the received signal's power. To measure the received power, a Tektronix RSA306 Spectrum Analyzer [11] is used.

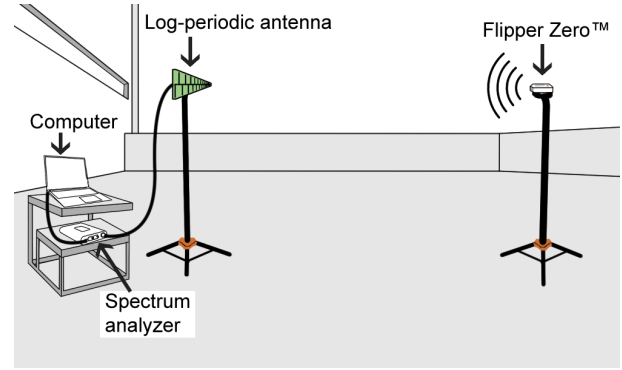


Fig. 1. Measured environment (line of sight).

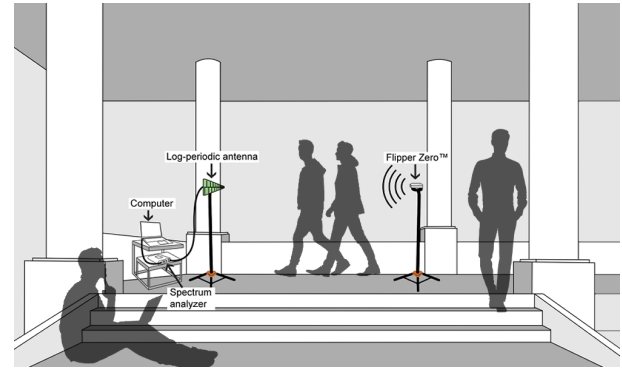


Fig. 2. Measured environment (student area).

It is worth mentioning that to distinguish the presence of a received signal's power, the knowledge of the noise power is needed. This can be calculated by replacing the antenna at the spectrum analyzer input with a matched load of 50 Ω and then collecting a set of noise samples every given time for a specific period [12],[13]. The obtained mean noise power was -106 dBm. Then, the center frequency of the spectrum analyzer is set to be equal to the pure tone signal, which is 925 MHz, with a span of 100 kHz. The receiver antenna is a log-periodic antenna, model Ettus LP0965, that works within the 800-6500 MHz range at 6 dBi gain [14]. The measuring station remains static, while the hacking device moves away one meter at a time from a reference distance $d_0 = 1$ meter up to a maximum of 15 meters. On each meter, five traces of 801 points samples are taken, hence the received power for each distance is the average power of the traces. The height from the floor to both the antenna and Flipper Zero™ is 1.30 meters.

C. Theoretical path loss model

Based on the empirical data, the log-distance path loss model can be found as follows

$$PL(\text{dB}) = PL(d_0) + 10n\log_{10}\left(\frac{d}{d_0}\right), \quad (1)$$

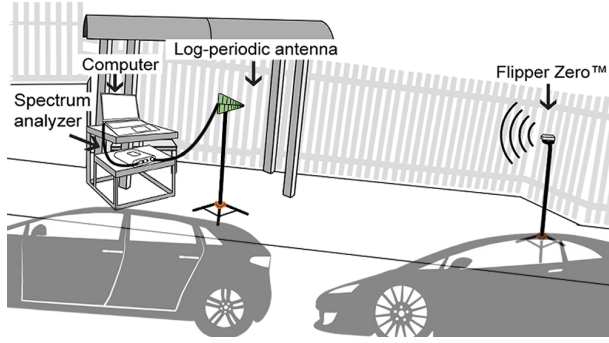


Fig. 3. Measured environment (sloping street).

where n is the path loss exponent obtained through a linear regression method with the taken power samples, d is the distance in meters and d_0 is a reference distance. A more accurate approach can be achieved by adding a zero-mean normal distributed fading factor X_σ [15]. This model is known as log-normal shadowing and is described as follows

$$PL(dB) = PL(d_0) + 10n\log_{10}\left(\frac{d}{d_0}\right) + X_\sigma. \quad (2)$$

Since X_σ is a random variable with Gaussian distribution, the probability that the received power P_r is above a certain value γ can be calculated with the Q-function [16] as

$$\text{Prob}[P_r(d) > \gamma] = Q\left(\frac{\gamma - \overline{P_r(d)}}{\sigma}\right), \quad (3)$$

where $\text{Prob}[\cdot]$ and $Q(\cdot)$ denote the probability operator and the Q-function operator, respectively.

IV. RESULTS

For all the scenarios, we have a set of 16 path loss samples from Flipper Zero™ sub-GHz transmitter taken at different distances, which we refer to as the empirical data. Such samples are used to solve a linear regression model to find a line whose slope ensures the smallest error ϵ , i.e., the line that minimizes the distance to the empirical data in the least squares sense. The slope of the line is the path loss exponent used in the theoretical path loss models, and ϵ is used to find the standard deviation of X_σ . Large n values mean that the path loss will increase at a higher rate, and the σ value is related to the dispersion of power values and how they deviate from the mean power.

The log-distance model shows a curve close to the empirical data points, whereas the log-normal shadowing model displays a computer-generated set of path loss values. Both models are valuable since they allow us to find the expected power at any distance d .

For the first scenario, in which there is a line-of-sight between transmitter and receiver, the results considering the empirical data samples, the linear regression model, the log-distance path loss model, and the log-normal shadowing

model, are presented in Fig. 4 with $n = 1.848$. Both theoretical path loss models display a total count of 141 points, i.e., an estimated value per 10 centimeters instead of a sampled value per meter, as in the empirical data.

The probability that the received power, P_r , is greater than a certain value γ is useful to predict the success of an attack given the proximity between the hacker and a legitimate IoT device. As previously mentioned, the minimum signal power requirements of a device depend on the technology it adopts and its sensitivity threshold. Therefore, knowing this probability of success is utilized to determine the physical area of vulnerability.

For scenario 1, from Fig. 5 we can obtain the distance at which the probability of reaching a required sensitivity level is at least 0.75. For example, with a probability of at least 0.75, the received signal at the IoT device receiver will be at least $\gamma = -50$ dBm at a maximum distance from the transmitter of less than a meter. For at least $\gamma = -60$ dBm the maximum distance can be a little more than 1 meter, and for $\gamma = -70$ dBm the maximum distance is less than 10 meters. Interestingly, it can be seen that, for a power level of at least $\gamma = -80$ dBm, which is enough for all the IoT devices presented in Table I to function properly, the maximum distance can be up to 15 meters with a probability of at least 0.75.

The empirical and theoretical data for the second scenario are shown in Fig. 6. It is clear that, due to external factors, both path loss models considered in this study increase at a higher rate when compared to scenario 1. Therefore, the same transmitter device will not perform the same way at the exact distances as the transmitted signal will be affected by other factors. The log-distance path loss model and the log-normal shadowing model are also displayed with 141 points and $n = 2.321$.

The maximum distance from which $\text{Prob}[P_r > \gamma] \geq 0.75$ is true, for γ of at least -60 dBm, is less than a meter. Similarly, for γ of at least -70 dBm, the maximum distance is less than 2 meters. For γ of at least -80 dBm, the maximum length is less than 5 meters, and for a γ of at least -90 dBm, it is less than 11 meters.

In the third scenario, the path loss is expected to be higher since more variables are out of control, such as the speed of vehicles and their dimensions, that affect the transmitted signal. In this study, we find $n = 2.556$, which is close enough to the value found in scenario 2. Figure 8 shows the empirical and theoretical path loss data with 16 and 141 sample points, respectively.

Lastly, the maximum distance in this scenario that satisfies $\text{Prob}[P_r > \gamma] \geq 0.75$ for γ of at least -60 dBm is less than a meter, for γ of at least -70 dBm, it is less than 2 meters. For a γ of at least -80 dBm, the maximum distance is less than 5 meters, whereas for a γ of at least -90 dBm, the maximum distance is less than 9 meters.

All the results are congruent with the obtained path loss, showing that the shorter the distance, the more likely the received power will exceed the threshold γ . Student area and

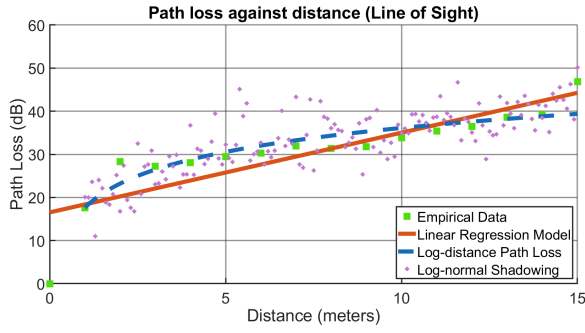


Fig. 4. Empirical data, linear regression model, average large-scale path loss, and log-normal shadowing with a line of sight (scenario 1).

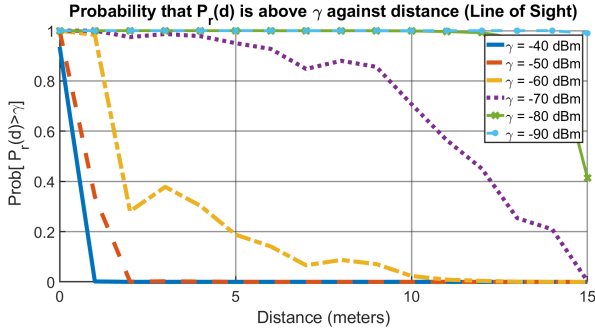


Fig. 5. Probability of P_r greater than γ with line of sight (scenario 1).

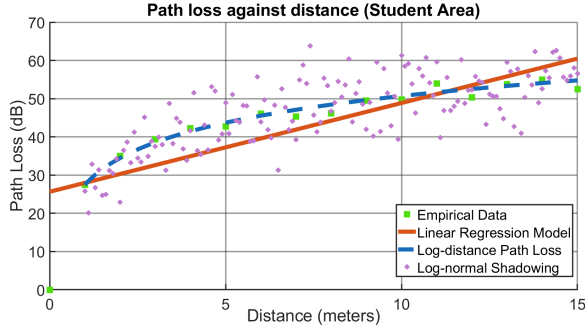


Fig. 6. Empirical data, linear regression model, average large-scale path loss and log-normal shadowing in a student area (scenario 2).

sloping street scenarios present smaller probabilities at the exact distances than direct line-of-sight scenarios.

The values of n for each scenario are summarized in Table II along with the standard deviation for each X_σ . Note that the path loss starts to increase as the distance between the measuring station and the transmitter increases. The lowest values of path loss are found when there is a direct line of sight, which represents the best scenario. Yet, it is more likely that attacks will be performed in other environments.

V. CONCLUSIONS

The paper analyzed the path loss experienced by the transmitter of a commercial hacking device in three different scenarios at the operating frequency of 925 MHz. The probability that the received power at some distance from the

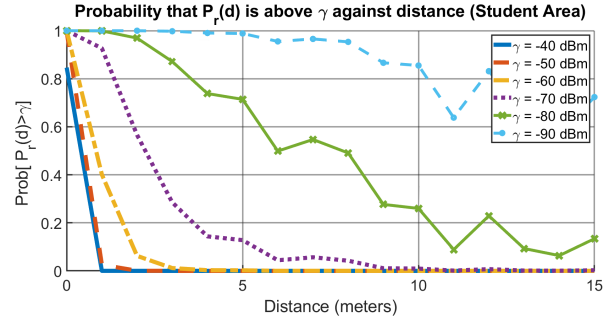


Fig. 7. Probability of P_r greater than γ in a student area (scenario 2).

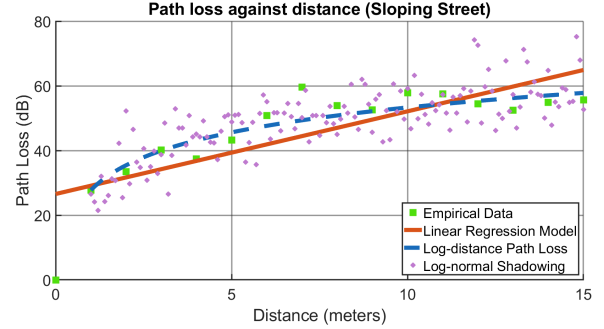


Fig. 8. Empirical data, linear regression model, average large-scale path loss, and log-normal shadowing in a sloping street (scenario 3).

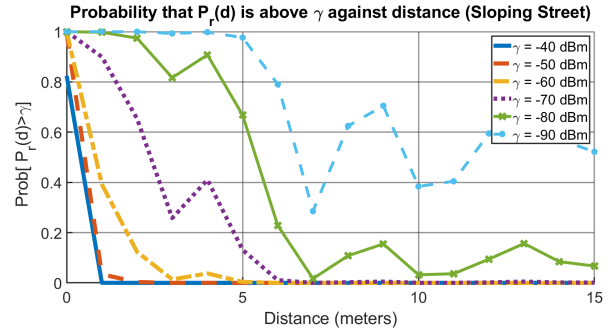


Fig. 9. Probability of P_r greater than γ in a sloped street (scenario 3).

transmitter is above a certain level has also been obtained for such three scenarios. Each scenario describes possible attack environments where a hacker could try to access IoT devices by sending fraudulent signals. What emerged is that ordinary users need to acknowledge the vulnerability of their systems since hacking tools are widely available. The success of an eavesdropper-transmitted signal meeting the power sensitivity

TABLE II
PATH LOSS EXPONENT AND STANDARD DEVIATION

Scenario	PL(d_0) [dB]	Path loss exponent (n)	σ
Line of sight	17.632	1.848	3.948
Student area	27.472	2.321	5.8762
Sloping street	27.760	2.556	6.4373

of the IoT receiver depends on several factors, such as the output power, the distance between both devices, the carrier wave, the modulation scheme, and others, as revealed by the conducted experiments. The results showed that using its lower transmission power, the Flipper Zero™ can reach IoT devices from distances up to 9 meters in the worst case of the analyzed scenarios and up to 15 meters when there is a line-of-sight between the transmitter and the receiver. These distances offer a reasonable range from which IoT devices can be hacked. A natural progression of this work is to perform more measurements at the other operating frequencies in which the Flipper Zero™ can work, for example, at 300-348 MHz, 387-464 MHz, and 779-928 MHz. Further studies are required to assess the potential of Flipper Zero™ as a hacking tool in different IoT contexts, more focusing on the IoT receiver response, which is the target of an eavesdropping attack.

ACKNOWLEDGMENT

All the experiments of this article were carried out with the equipment of the Coordination for Innovation and Application of Science and Technology (CIACYT) of the Autonomous University of San Luis Potosí (UASLP).

REFERENCES

- [1] Al-Fuqaha, M. Guizani, M. Mohammadi, M. Aledhari, and M. Ayyash, "Internet of Things: A Survey on Enabling Technologies, Protocols, and Applications," in *IEEE Communications Surveys & Tutorials*, vol. 17, no. 4, pp. 2347-2376, fourth quarter 2015, doi: 10.1109/COMST.2015.2444095.
- [2] M. Aqeel, F. Ali, M. W. Iqbal, T. A. Rana, M. Arif, Md. R. Auwal, "A Review of Security and Privacy Concerns in the Internet of Things (IoT)," *Journal of Sensors*, vol. 2022, third quarter 2022, doi: 10.1155/2022/5724168.
- [3] F. Hussain, R. Hussain, S. A. Hassan and E. Hossain, "Machine Learning in IoT Security: Current Solutions and Future Challenges," in *IEEE Communications Surveys & Tutorials*, vol. 22, no. 3, pp. 1686-1721, third quarter 2020, doi: 10.1109/COMST.2020.2986444.
- [4] M. A. Al-Garadi, A. Mohamed, A. K. Al-Ali, X. Du, I. Ali, and M. Guizani, "A Survey of Machine and Deep Learning Methods for Internet of Things (IoT) Security," in *IEEE Communications Surveys & Tutorials*, vol. 22, no. 3, pp. 1646-1685, third quarter 2020, doi: 10.1109/COMST.2020.2988293.
- [5] A. Murzaeva, B. Kepçeoğlu and S. Demirci, "Survey of Network Security Issues and Solutions for the IoT," *3rd International Symposium on Multidisciplinary Studies and Innovative Technologies (ISMSIT)*, Ankara, Turkey, 2019, pp. 1-6, doi: 10.1109/ISMSIT.2019.8932957.
- [6] A. Kumbhar, "Overview of ISM Bands and Software-Defined Radio Experimentation," in *Wireless Pers Commun*, vol. 97, no. 3, pp. 3743-3756, Dec. 2017, doi:10.1007/s11277-017-4696-z.
- [7] D. B. Rawat, G. Yan, B. B. Bista, and V. Chandra, "Wireless Network Security: An Overview," in *Building Next-Generation Converged Networks: Theory and Practice*, 1st. ed. Boca Raton, FL: CRC Press Taylor & Francis Group, 2013, ch. 7, pp. 200-219.
- [8] E. M. Ar-Reyouchi, K. Ghomid, R. Yahiaoui and O. Elmazria, "Optimized Reception Sensitivity of WBAN Sensors Exploiting Network Coding and Modulation Techniques in an Advanced NB-IoT," in *IEEE Access*, vol. 10, pp. 35784-35794, 2022, doi: 10.1109/ACCESS.2022.3163314.
- [9] Flipper Zero, "Sub-ghz." docs.flipperzero.one. <https://docs.flipperzero.one/sub-ghz> (accessed May 20, 2023).
- [10] Texas Instruments, "Low-Power Sub-1 GHz RF Transceiver," CC1101 datasheet, Nov. 2013 [Revised May 2023].
- [11] Tektronix, "RSA306 USB Real Time Spectrum Analyzer Datasheet," RSA306 datasheet, Dec. 2015 [Revised June 2023].
- [12] M. López-Benítez and F. Casadevall, "Methodological aspects of spectrum occupancy evaluation in the context of cognitive radio," in *European Transactions on Telecommunications*, vol. 21, pp. 680 - 693, Nov. 2010, doi: 10.1002/ett.1453.
- [13] L. Lopez-Lopez, M. Cardenas-Juarez, E. Stevens-Navarro, U. Pineda-Rico, A. Arce and A. G. Orozco-Lugo, "Superimposed Training Combined Approach for a Reduced Phase of Spectrum Sensing in Cognitive Radio" in *Sensors*, vol. 19, pp. 2425, May 2019, doi: 10.3390/s19112425.
- [14] Ettus, "850-6500 MHz PCB LP WA5VJB," LP0965 datasheet, Jan. 2012 [Revised June 2023].
- [15] J. Frolik, "Implementation of Handheld, RF Test Equipment in the Classroom and the Field," in *IEEE Transactions on Education*, vol. 50, no. 3, pp. 182-187, Aug. 2007, doi: 10.1109/TE.2007.900022.
- [16] T. S. Rappaport, "Mobile Radio Propagation: Large-Scale Path Loss," in *Wireless Communications: Principles and practice*, 2nd ed, Upper Saddle River, NJ: Prentice Hall PTR, 2009, pp. 102-110.